

JOINS IN SQL

1) *Creating a table and inserting the values:*

```
mysql> CREATE DATABASE IF NOT EXISTS CARSHOWROOM ;
Query OK, 1 row affected (0.01 sec)

mysql> USE CARSHOWROOM;
Database changed
mysql> CREATE TABLE CARS (
  ->   CarID INT PRIMARY KEY,
  ->   Brand VARCHAR(50),
  ->   Model VARCHAR(50),
  ->   YearManufacture INT,
  ->   Price DECIMAL(10, 2)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE SALES (
  ->   InvoiceNo INT PRIMARY KEY,
  ->   CarID INT,
  ->   CustomerName VARCHAR(100),
  ->   SaleDate DATE,
  ->   SellingPrice DECIMAL(10, 2),
  ->   FOREIGN KEY (CarID) REFERENCES CARS(CarID)
  -> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO CARS (CarID, Brand, Model, YearManufacture, Price) VALUES
  -> (101, 'Tesla', 'Model 3', 2024, 40000.00),
  -> (102, 'Ford', 'Mustang Mustang GT', 2023, 55000.00),
  -> (103, 'Toyota', 'RAV4 Hybrid', 2024, 32000.00),
  -> (104, 'BMW', 'i4 M50', 2025, 67000.00);
Query OK, 4 rows affected (0.01 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> INSERT INTO SALES (InvoiceNo, CarID, CustomerName, SaleDate, SellingPrice) VALUES
  -> (5001, 101, 'Alice Johnson', '2026-03-15', 39500.00),
  -> (5002, 103, 'Bob Smith', '2026-04-02', 32000.00),
  -> (5003, 101, 'Charlie Brown', '2026-05-20', 40000.00);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO SALES (InvoiceNo, CarID, CustomerName, SaleDate, SellingPrice)
  -> VALUES (5004, 104, 'Mohith Gowda', '2026-09-22', 67000.00);
Query OK, 1 row affected (0.01 sec)
```

2) Join:

```
mysql> SELECT
  ->     S.InvoiceNo,
  ->     S.CustomerName,
  ->     C.Brand,
  ->     C.Model,
  ->     S.SellingPrice,
  ->     S.SaleDate
  -> FROM SALES S
  -> INNER JOIN CARS C ON S.CarID = C.CarID;
```

InvoiceNo	CustomerName	Brand	Model	SellingPrice	SaleDate
5001	Alice Johnson	Tesla	Model 3	39500.00	2026-03-15
5002	Bob Smith	Toyota	RAV4 Hybrid	32000.00	2026-04-02
5003	Charlie Brown	Tesla	Model 3	40000.00	2026-05-20
5004	Mohith Gowda	BMW	i4 M50	67000.00	2026-09-22

4 rows in set (0.00 sec)

3) Left Join:

```
mysql> INSERT INTO CARS (CarID, Brand, Model, YearManufacture, Price)
  -> VALUES (105, 'Porsche', 'Carrera', 2026, 120000.00);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT
  ->     C.CarID,
  ->     C.Brand,
  ->     C.Model,
  ->     S.InvoiceNo,
  ->     S.CustomerName,
  ->     S.SellingPrice
  -> FROM CARS C
  -> LEFT JOIN SALES S ON C.CarID = S.CarID;
```

CarID	Brand	Model	InvoiceNo	CustomerName	SellingPrice
101	Tesla	Model 3	5001	Alice Johnson	39500.00
101	Tesla	Model 3	5003	Charlie Brown	40000.00
102	Ford	Mustang Mustang GT	NULL	NULL	NULL
103	Toyota	RAV4 Hybrid	5002	Bob Smith	32000.00
104	BMW	i4 M50	5004	Mohith Gowda	67000.00
105	Porsche	Carrera	NULL	NULL	NULL

6 rows in set (0.00 sec)

4) Right Join:

```
mysql> SELECT
->     S.InvoiceNo,
->     S.CustomerName,
->     S.SellingPrice,
->     C.Brand,
->     C.Model
-> FROM CARS C
-> RIGHT JOIN SALES S ON C.CarID = S.CarID;
```

InvoiceNo	CustomerName	SellingPrice	Brand	Model
5001	Alice Johnson	39500.00	Tesla	Model 3
5002	Bob Smith	32000.00	Toyota	RAV4 Hybrid
5003	Charlie Brown	40000.00	Tesla	Model 3
5004	Mohith Gowda	67000.00	BMW	i4 M50

4 rows in set (0.00 sec)

5) Inner Join:

```
mysql> SELECT
->     S.InvoiceNo,
->     S.CustomerName,
->     C.Brand,
->     C.Model,
->     S.SellingPrice,
->     S.SaleDate
-> FROM SALES S
-> INNER JOIN CARS C ON S.CarID = C.CarID;
```

InvoiceNo	CustomerName	Brand	Model	SellingPrice	SaleDate
5001	Alice Johnson	Tesla	Model 3	39500.00	2026-03-15
5002	Bob Smith	Toyota	RAV4 Hybrid	32000.00	2026-04-02
5003	Charlie Brown	Tesla	Model 3	40000.00	2026-05-20
5004	Mohith Gowda	BMW	i4 M50	67000.00	2026-09-22

4 rows in set (0.00 sec)

6) Full Outer Join:

```
mysql> SELECT C.CarID, C.Brand, C.Model, S.InvoiceNo, S.CustomerName, S.SellingPrice
-> FROM CARS C
-> LEFT JOIN SALES S ON C.CarID = S.CarID
->
-> UNION
->
-> SELECT C.CarID, C.Brand, C.Model, S.InvoiceNo, S.CustomerName, S.SellingPrice
-> FROM CARS C
-> RIGHT JOIN SALES S ON C.CarID = S.CarID;
```

CarID	Brand	Model	InvoiceNo	CustomerName	SellingPrice
101	Tesla	Model 3	5003	Charlie Brown	40000.00
101	Tesla	Model 3	5001	Alice Johnson	39500.00
102	Ford	Mustang Mustang GT	NULL	NULL	NULL
103	Toyota	RAV4 Hybrid	5002	Bob Smith	32000.00
104	BMW	i4 M50	5004	Mohith Gowda	67000.00
105	Porsche	Carrera	NULL	NULL	NULL

6 rows in set (0.01 sec)